Haoxuan Cheng

Curriculum Vitae

School of Mathematical Sciences, Fudan University
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Education

Fudan University — Direct-entry PhD program in mathematics, 2025–present.

South China Normal University — Previous studies in mathematics.

Research interests

Discrete geometric analysis; Ricci curvature on graphs; spectral graph theory; geometric analysis on manifolds, including volume growth, stochastic processes, and curvature flows.

Support

Class B funding, National High-Level Talent Training Center at Fudan University.

Talks and poster presentations

- **Discrete Einstein Metrics on Trees.** Poster presentation, *Geometric and analytic functional inequalities on discrete spaces*, Tsinghua Sanya International Mathematics Forum, Sanya, China. Conference: 14–18 September 2026. Conference page.
- **Monotonicity and Limiting Behavior of Einstein Trees I: Leaf Attachment.** Beijing Institute of Mathematical Sciences and Applications (BIMSA), 26 June 2026. Talk page.
- **A Classification of Positive-Curvature Discrete Einstein Metrics on Trees.** BIMSA, 20 May 2026. Seminar listing.

AI and machine learning projects

- **Graph Geometry Research Workspace (AI4M).** August 2026–present. Independently designed and open-sourced a workbench for AI-assisted mathematical research, tracking proof attempts, dependencies, and computational evidence. Current explorations include clearer mathematical writing and support for understanding definitions, ideas, and proofs. Public repository.

- **OCT foveal layer localization and thickness measurement.** 2026. Second inventor; patent application pending. Designed keypoint detection and boundary refinement for the outer nuclear layer. With model weights fixed, refinement reduced upper-boundary vertical MAE from 0.826 to 0.331 pixels on 119 manually annotated validation images.
- **LLM-assisted dividend forecasting.** 2026. Commissioned technical collaboration. Developed a LangGraph workflow combining evidence-linked hypotheses from earnings call transcripts with rule-based, Holt exponential smoothing, and ridge-regression baselines. Collaborated with business analysts on evaluation across 50 companies and iterative improvements.
- **Transfer learning for atrial fibrillation recognition.** March 2023–April 2024. Contributed wavelet-based ECG preprocessing and the research report to a project using convolutional networks and transfer learning on MIT-BIH data, adapting single-lead recognition to dual-lead signals.

Technical experience

AI-assisted development with Codex; prompt and context engineering; model APIs and structured outputs; agent workflows and tool calling; baseline comparisons and case analysis. Practical experience with PyTorch, transfer learning, keypoint detection, and LangGraph.

AI and machine learning projects updated 9 November 2026; preprint list last updated 6 October 2026.

Preprints

10. **A counterexample to Yau's bounded-mean-curvature embedding question**
    Haoxuan Cheng. 2026. arXiv:2610.00153.

9. **Umbilic slopes and cubic Weingarten surfaces**
    Haoxuan Cheng. 2026. arXiv:2609.19188.

8. **Bounded curvature manifolds without Euclidean isometric immersions of bounded mean curvature**
    Haoxuan Cheng. 2026. arXiv:2609.19168.

7. **Unbounded normalized scalar curvature integrals in dimension four**
    Haoxuan Cheng. 2026. arXiv:2609.10323.

6. **Singular Rotational Self-Similar Tori for Odd σ_k-Curvature Flows**
    Haoxuan Cheng, Junqi Lai, Guoxin Wei. 2026. arXiv:2609.01346.

5. **Volume Growth and Recurrence of Fractional Powers of the Laplace--Beltrami Operator**
    Haoxuan Cheng. 2026. arXiv:2608.26509.

4. **Edge Subdivision and the Perron Eigenvalue of Tree Ricci Matrices**
    Shuliang Bai, Haoxuan Cheng, Bobo Hua. 2026. arXiv:2605.30949.

3. **Spectral Monotonicity under Leaf Attachment and Limiting Behavior in Discrete Einstein Trees**
    Shuliang Bai, Haoxuan Cheng, Bobo Hua. 2026. arXiv:2605.23379.

2. **Positive-Curvature Discrete Einstein Metrics on Trees**
    Haoxuan Cheng. 2026. arXiv:2605.20862.

1. **Discrete Einstein metrics on trees**
    Shuliang Bai, Haoxuan Cheng, Bobo Hua. 2026. arXiv:2604.22449.